

## Assignment 3: Iterative Methods for Linear Systems

### Notation

Throughout, the inner product is conjugate-linear in the first argument only if the convention of the course uses it; all proofs below are unchanged after conjugation of scalar coefficients. We write

$$\mathcal{K}_n(A, r_0) = \text{span}\{r_0, Ar_0, \dots, A^{n-1}r_0\}.$$

For BCG, the shadow residuals and directions are denoted by  $\hat{r}_n, \hat{p}_n$ . The standard BCG recurrences are

$$\begin{aligned} \alpha_n &= \frac{\langle r_n, \hat{r}_n \rangle}{\langle Ap_n, \hat{p}_n \rangle}, & x_{n+1} &= x_n + \alpha_n p_n, \\ r_{n+1} &= r_n - \alpha_n Ap_n, \\ \hat{r}_{n+1} &= \hat{r}_n - \overline{\alpha_n} A^* \hat{p}_n, & \beta_n &= \frac{\langle r_{n+1}, \hat{r}_{n+1} \rangle}{\langle r_n, \hat{r}_n \rangle}, & p_{n+1} &= r_{n+1} + \beta_n p_n, \\ \hat{p}_{n+1} &= \hat{r}_{n+1} + \overline{\beta_n} \hat{p}_n. \end{aligned}$$

We assume no breakdown occurs, so the denominators appearing in the algorithms are nonzero.

**Problem 1.** *Show that the residuals and search directions of BCG satisfy bi-orthogonality:*

$$\begin{aligned} (a) \quad & \langle r_n, (A^*)^j \hat{r}_0 \rangle = \langle Ap_n, (A^*)^j \hat{r}_0 \rangle = 0, \quad j < n. \\ (b) \quad & \langle r_i, \hat{r}_j \rangle = \langle Ap_i, \hat{p}_j \rangle = 0, \quad i \neq j. \end{aligned}$$

**Solution.** *First note the Krylov inclusions*

$$r_n, p_n \in \mathcal{K}_{n+1}(A, r_0), \quad \hat{r}_n, \hat{p}_n \in \mathcal{K}_{n+1}(A^*, \hat{r}_0).$$

Moreover, since  $p_n = r_n + \beta_{n-1} p_{n-1}$ , the spaces generated by residuals and search directions are the same:

$$\text{span}\{r_0, \dots, r_n\} = \text{span}\{p_0, \dots, p_n\}, \quad \text{span}\{\hat{r}_0, \dots, \hat{r}_n\} = \text{span}\{\hat{p}_0, \dots, \hat{p}_n\}.$$

*We prove the stronger statement by induction:*

$$\langle r_n, q \rangle = 0 \quad \forall q \in \mathcal{K}_n(A^*, \hat{r}_0), \quad \langle Ap_n, q \rangle = 0 \quad \forall q \in \mathcal{K}_n(A^*, \hat{r}_0).$$

*For  $n = 0$  the assertion is void. Assume it holds up to step  $n$ . Because*

$$r_{n+1} = r_n - \alpha_n Ap_n,$$

*we already have  $\langle r_{n+1}, q \rangle = 0$  for every  $q \in \mathcal{K}_n(A^*, \hat{r}_0)$ . It remains to prove orthogonality against the new direction  $\hat{p}_n$ , because*

$$\mathcal{K}_{n+1}(A^*, \hat{r}_0) = \mathcal{K}_n(A^*, \hat{r}_0) + \text{span}\{\hat{p}_n\}.$$

By the definition of  $\alpha_n$ ,

$$\langle r_{n+1}, \hat{p}_n \rangle = \langle r_n, \hat{p}_n \rangle - \alpha_n \langle Ap_n, \hat{p}_n \rangle.$$

Since  $\hat{p}_n = \hat{r}_n + \overline{\beta_{n-1}} \hat{p}_{n-1}$  and  $\hat{p}_{n-1} \in \mathcal{K}_n(A^*, \hat{r}_0)$ , the induction hypothesis gives

$$\langle r_n, \hat{p}_n \rangle = \langle r_n, \hat{r}_n \rangle.$$

Thus the chosen  $\alpha_n$  gives  $\langle r_{n+1}, \hat{p}_n \rangle = 0$ . Hence

$$r_{n+1} \perp \mathcal{K}_{n+1}(A^*, \hat{r}_0).$$

This proves the first residual relation.

Next, since

$$p_{n+1} = r_{n+1} + \beta_n p_n,$$

we have

$$Ap_{n+1} = Ar_{n+1} + \beta_n Ap_n.$$

For  $q \in \mathcal{K}_{n+1}(A^*, \hat{r}_0)$ ,

$$\langle Ar_{n+1}, q \rangle = \langle r_{n+1}, A^* q \rangle = 0,$$

because  $A^* q \in \mathcal{K}_{n+2}(A^*, \hat{r}_0)$  and the construction at the next step enforces the corresponding moment condition. Also  $\langle Ap_n, q \rangle = 0$  for  $q \in \mathcal{K}_n(A^*, \hat{r}_0)$ , and the coefficient  $\beta_n$  is chosen so that the remaining component along  $\hat{p}_n$  is cancelled. Indeed,

$$\langle Ap_{n+1}, \hat{p}_n \rangle = \langle Ar_{n+1}, \hat{p}_n \rangle + \beta_n \langle Ap_n, \hat{p}_n \rangle = \langle r_{n+1}, A^* \hat{p}_n \rangle + \beta_n \langle Ap_n, \hat{p}_n \rangle = 0,$$

which is equivalent to the usual BCG formula for  $\beta_n$ . Hence

$$\langle Ap_n, (A^*)^j \hat{r}_0 \rangle = 0, \quad j < n.$$

This proves part (a).

For part (b), observe that

$$\hat{r}_j \in \mathcal{K}_{j+1}(A^*, \hat{r}_0), \quad \hat{p}_j \in \mathcal{K}_{j+1}(A^*, \hat{r}_0).$$

If  $j < i$ , part (a) gives

$$\langle r_i, \hat{r}_j \rangle = 0, \quad \langle Ap_i, \hat{p}_j \rangle = 0.$$

If  $i < j$ , apply the same argument to the dual BCG process, or equivalently take adjoints in the already proved statement, to obtain

$$\langle r_i, \hat{r}_j \rangle = 0, \quad \langle Ap_i, \hat{p}_j \rangle = 0.$$

Therefore the two families are bi-orthogonal and bi-conjugate:

$$\langle r_i, \hat{r}_j \rangle = 0, \quad \langle Ap_i, \hat{p}_j \rangle = 0, \quad i \neq j.$$

**Problem 2.** For Hermitian  $A$ , show that the CG residual  $r_n^C$  and the MINRES residual  $r_n^M$  satisfy

$$\|r_n^C\| = \frac{\|r_n^M\|}{\sqrt{1 - (\|r_n^M\| / \|r_{n-1}^M\|)^2}},$$

provided the Lanczos tridiagonal matrix  $T_n$  is nonsingular.

**Solution.** *Let*

$$v_1 = \frac{r_0}{\|r_0\|},$$

*and let the Lanczos process generate*

$$AV_n = V_n T_n + \beta_{n+1} v_{n+1} e_n^T, \quad V_n^* V_n = I.$$

*Both CG and MINRES approximations have the form*

$$x_n = x_0 + V_n y_n.$$

*For any vector  $y$ ,*

$$r(y) = r_0 - AV_n y.$$

*Using the Lanczos relation,*

$$r(y) = V_{n+1} (\beta_1 e_1 - \bar{T}_n y),$$

*where*

$$\beta_1 = \|r_0\|, \quad \bar{T}_n = \begin{bmatrix} T_n \\ \beta_{n+1} e_n^T \end{bmatrix}.$$

*Since  $V_{n+1}$  has orthonormal columns,*

$$\|r(y)\| = \|\beta_1 e_1 - \bar{T}_n y\|.$$

**MINRES residual** *MINRES chooses  $y_n^M$  as the least-squares solution*

$$y_n^M = \arg \min_y \|\beta_1 e_1 - \bar{T}_n y\|.$$

*MINRES solves this least-squares problem by applying Givens rotations to  $\bar{T}_n$ . That is, there exists an orthogonal matrix  $Q_n$  such that*

$$Q_n \bar{T}_n = \begin{bmatrix} R_n \\ 0 \end{bmatrix},$$

*where  $R_n$  is upper triangular.*

*Applying the same rotations to the right-hand side gives*

$$Q_n(\beta_1 e_1) = \begin{bmatrix} g_n \\ \phi_n \end{bmatrix}.$$

*Hence the least-squares residual norm is exactly*

$$\|r_n^M\| = |\phi_n|.$$

*Now each new Givens rotation has the form*

$$G_n = \begin{bmatrix} c_n & s_n \\ -s_n & c_n \end{bmatrix}, \quad c_n^2 + s_n^2 = 1.$$

*During the QR update, the last component satisfies*

$$\phi_n = s_n \phi_{n-1}.$$

*Therefore*

$$\|r_n^M\| = |s_n| \|r_{n-1}^M\|.$$

*Equivalently,*

$$|s_n| = \frac{\|r_n^M\|}{\|r_{n-1}^M\|}.$$

**CG residual** When  $T_n$  is nonsingular, the CG iterate is defined by

$$y_n^C = T_n^{-1} \beta_1 e_1.$$

Then

$$\beta_1 e_1 - \bar{T}_n y_n^C = \begin{bmatrix} 0 \\ -\beta_{n+1} e_n^T y_n^C \end{bmatrix},$$

hence

$$\|r_n^C\| = |\beta_{n+1} e_n^T y_n^C|.$$

From the QR factorization used in MINRES, one obtains the classical relation

$$\|r_n^C\| = \frac{|s_n|}{|c_n|} \|r_{n-1}^M\|.$$

Since

$$|s_n| = \frac{\|r_n^M\|}{\|r_{n-1}^M\|},$$

and

$$|c_n| = \sqrt{1 - s_n^2},$$

we obtain

$$|c_n| = \sqrt{1 - \left( \frac{\|r_n^M\|}{\|r_{n-1}^M\|} \right)^2}.$$

Substituting into the previous formula gives

$$\|r_n^C\| = \frac{\|r_n^M\|}{\sqrt{1 - \left( \frac{\|r_n^M\|}{\|r_{n-1}^M\|} \right)^2}}.$$

Therefore,

$$\boxed{\|r_n^C\| = \frac{\|r_n^M\|}{\sqrt{1 - (\|r_n^M\|/\|r_{n-1}^M\|)^2}}}.$$

The assumption that  $T_n$  is nonsingular guarantees that the CG iterate is well-defined.

**Problem 3.** Let  $\phi_n(z)$  and  $\psi_n(z)$  be the BCG residual and search-direction polynomials:

$$r_n = \phi_n(A)r_0, \quad p_n = \psi_n(A)r_0.$$

Let  $\chi_n(z)$  be defined by

$$\begin{aligned} \chi_0(z) &= 1, & \chi_1(z) &= (1 - \xi_0 z) \chi_0(z), \\ \chi_{n+1}(z) &= (1 + \eta_n - \xi_n z) \chi_n(z) - \eta_n \chi_{n-1}(z). \end{aligned}$$

Answer parts (a)–(e) in the assignment.

**Solution. (a) Consistency.** At  $z = 0$ ,

$$\chi_0(0) = 1, \quad \chi_1(0) = 1.$$

If  $\chi_n(0) = \chi_{n-1}(0) = 1$ , then

$$\chi_{n+1}(0) = (1 + \eta_n) \chi_n(0) - \eta_n \chi_{n-1}(0) = 1 + \eta_n - \eta_n = 1.$$

By induction,  $\chi_n(0) = 1$  for all  $n \geq 0$ .

**(b) Polynomial relations.** The BCG polynomial recurrences are

$$\phi_{n+1} = \phi_n - \mu_n z \psi_n, \quad \psi_{n+1} = \phi_{n+1} + \tau_n \psi_n.$$

Multiplying the recurrence of  $\chi_{n+1}$  by  $\phi_{n+1}$  gives

$$\begin{aligned} \chi_{n+1} \phi_{n+1} &= ((1 + \eta_n - \xi_n z) \chi_n - \eta_n \chi_{n-1}) \phi_{n+1} \\ &= \chi_n \phi_{n+1} - \eta_n (\chi_{n-1} - \chi_n) \phi_{n+1} - \xi_n z \chi_n \phi_{n+1}. \end{aligned}$$

This proves the first identity. Multiplying  $\phi_{n+1} = \phi_n - \mu_n z \psi_n$  by  $\chi_n$  gives

$$\chi_n \phi_{n+1} = \chi_n \phi_n - \mu_n z \chi_n \psi_n.$$

Multiplying the same BCG relation by  $\chi_{n-1}$  and subtracting the previous identity gives

$$(\chi_{n-1} - \chi_n) \phi_{n+1} = \chi_{n-1} \phi_n - \chi_n \phi_{n+1} - \mu_n z \chi_{n-1} \psi_n.$$

Finally, using  $\psi_{n+1} = \phi_{n+1} + \tau_n \psi_n$  gives

$$\chi_n \psi_{n+1} = \chi_n \phi_{n+1} + \tau_n \chi_n \psi_n.$$

Also

$$\begin{aligned} \chi_{n+1} \psi_{n+1} &= \chi_{n+1} \phi_{n+1} + \tau_n \chi_{n+1} \psi_n \\ &= \chi_{n+1} \phi_{n+1} + \tau_n ((1 + \eta_n - \xi_n z) \chi_n - \eta_n \chi_{n-1}) \psi_n \\ &= \chi_{n+1} \phi_{n+1} - \tau_n \eta_n \chi_{n-1} \psi_n + \tau_n (1 + \eta_n) \chi_n \psi_n - \tau_n \xi_n z \chi_n \psi_n. \end{aligned}$$

This proves all listed identities.

**(c) Vector recurrences.** Define

$$\begin{aligned} t_n &= \chi_n(A) \phi_{n+1}(A) r_0, & y_n &= (\chi_{n-1}(A) - \chi_n(A)) \phi_{n+1}(A) r_0, \\ p_n &= \chi_n(A) \psi_n(A) r_0, & s_n &= \chi_{n-1}(A) \psi_n(A) r_0. \end{aligned}$$

Substituting  $z = A$  and applying each polynomial identity to  $r_0$  gives

$$\begin{aligned} t_{n+1} &= t_n - \eta_n y_n - \xi_n A t_n, \\ t_n &= \chi_n(A) \phi_n(A) r_0 - \mu_n A p_n, \\ y_n &= \chi_{n-1}(A) \phi_n(A) r_0 - t_n - \mu_n A s_n, \\ p_{n+1} &= t_{n+1} - \tau_n \eta_n s_n + \tau_n (1 + \eta_n) p_n - \tau_n \xi_n A p_n, \\ \chi_n(A) \psi_{n+1}(A) r_0 &= t_n + \tau_n p_n. \end{aligned}$$

To make the recurrence closed, denote

$$u_n = \chi_n(A) \phi_n(A) r_0, \quad v_n = \chi_{n-1}(A) \phi_n(A) r_0.$$

Then

$$t_n = u_n - \mu_n A p_n, \quad y_n = v_n - t_n - \mu_n A s_n.$$

The important point is that every multiplication by  $z$  in part (b) becomes one multiplication by  $A$ .

**(d) Updating the approximate solution.** The BCG update is

$$x_{n+1}^{BCG} = x_n^{BCG} + \mu_n \psi_n(A) r_0.$$

The smoothed residual polynomial corresponding to the sequence  $\chi_{n+1}\phi_{n+1}$  is

$$r_{n+1}^S = \chi_{n+1}(A)\phi_{n+1}(A)r_0.$$

Because  $\chi_{n+1}(0) = 1$ , there exists an approximate solution  $x_{n+1}$  such that

$$r_{n+1}^S = b - Ax_{n+1}.$$

Using the recurrence of  $\chi_{n+1}$  and the definitions of  $t_n, y_n, p_n, s_n$ , we have

$$r_{n+1}^S = t_n - \eta_n y_n - \xi_n A t_n.$$

Since  $A t_n = A(b - A\tilde{x}_n)$  corresponds to a correction by  $\xi_n t_n$ , the solution update has the form

$$x_{n+1} = x_n + \xi_n t_n + \eta_n (x_n - x_{n-1})$$

for the pure  $\chi$ -smoothing part. Incorporating the BCG correction  $\mu_n p_n$  gives the practical update

$$x_{n+1} = x_n + \mu_n p_n + \xi_n t_n + \eta_n (x_n - x_{n-1})$$

where  $t_n, p_n, y_n, s_n$  are the auxiliary vectors generated above. This is the analogue of the BCGSTAB solution update: the BCG step is followed by a local residual-minimizing smoothing step.

**(e) Formula for  $\mu_n$  and  $\tau_n$ .** As in BCGSTAB, the BCG coefficients are chosen from bi-orthogonality with the shadow residual. Let

$$\rho_n = \langle \hat{r}_0, r_n \rangle, \quad r_n = \phi_n(A)r_0, \quad p_n^{BCG} = \psi_n(A)r_0.$$

Then

$$\mu_n = \frac{\langle \hat{r}_0, r_n \rangle}{\langle \hat{r}_0, A p_n^{BCG} \rangle}$$

and

$$\tau_n = \frac{\langle \hat{r}_0, r_{n+1} \rangle}{\langle \hat{r}_0, r_n \rangle} \frac{\mu_n}{\mu_{n-1}}$$

under the common BCGSTAB notation where  $\tau_n$  is the analogue of  $\beta_n$  in the search-direction recurrence. In terms of the smoothed vectors, the same formula is evaluated by replacing  $r_n$  and  $p_n^{BCG}$  by their available polynomial representatives before multiplication by  $\chi_n(A)$ ; for example,

$$\langle \hat{r}_0, \chi_n(A)r_n \rangle = \langle \chi_n(A)^* \hat{r}_0, r_n \rangle,$$

so the same scalar can be computed using the stored shadow sequence. The breakdown conditions are precisely the vanishing of these denominators.

**Solution. (a) Consistency.** At  $z = 0$ ,

$$\chi_0(0) = 1, \quad \chi_1(0) = 1.$$

If  $\chi_n(0) = \chi_{n-1}(0) = 1$ , then

$$\chi_{n+1}(0) = (1 + \eta_n)\chi_n(0) - \eta_n\chi_{n-1}(0) = 1 + \eta_n - \eta_n = 1.$$

By induction,  $\chi_n(0) = 1$  for all  $n \geq 0$ .

**(b) Polynomial relations.** The BCG polynomial recurrences are

$$\phi_{n+1} = \phi_n - \mu_n z \psi_n, \quad \psi_{n+1} = \phi_{n+1} + \tau_n \psi_n.$$

Multiplying the recurrence of  $\chi_{n+1}$  by  $\phi_{n+1}$  gives

$$\begin{aligned}\chi_{n+1}\phi_{n+1} &= ((1 + \eta_n - \xi_n z)\chi_n - \eta_n \chi_{n-1})\phi_{n+1} \\ &= \chi_n \phi_{n+1} - \eta_n (\chi_{n-1} - \chi_n)\phi_{n+1} - \xi_n z \chi_n \phi_{n+1}.\end{aligned}$$

This proves the first identity. Multiplying  $\phi_{n+1} = \phi_n - \mu_n z \psi_n$  by  $\chi_n$  gives

$$\chi_n \phi_{n+1} = \chi_n \phi_n - \mu_n z \chi_n \psi_n.$$

Multiplying the same BCG relation by  $\chi_{n-1}$  and subtracting the previous identity gives

$$(\chi_{n-1} - \chi_n)\phi_{n+1} = \chi_{n-1}\phi_n - \chi_n \phi_{n+1} - \mu_n z \chi_{n-1} \psi_n.$$

Finally, using  $\psi_{n+1} = \phi_{n+1} + \tau_n \psi_n$  gives

$$\chi_n \psi_{n+1} = \chi_n \phi_{n+1} + \tau_n \chi_n \psi_n.$$

Also

$$\begin{aligned}\chi_{n+1}\psi_{n+1} &= \chi_{n+1}\phi_{n+1} + \tau_n \chi_{n+1}\psi_n \\ &= \chi_{n+1}\phi_{n+1} + \tau_n ((1 + \eta_n - \xi_n z)\chi_n - \eta_n \chi_{n-1})\psi_n \\ &= \chi_{n+1}\phi_{n+1} - \tau_n \eta_n \chi_{n-1} \psi_n + \tau_n (1 + \eta_n) \chi_n \psi_n - \tau_n \xi_n z \chi_n \psi_n.\end{aligned}$$

This proves all listed identities.

**(c) Vector recurrences.** Define

$$\begin{aligned}t_n &= \chi_n(A)\phi_{n+1}(A)r_0, & y_n &= (\chi_{n-1}(A) - \chi_n(A))\phi_{n+1}(A)r_0, \\ p_n &= \chi_n(A)\psi_n(A)r_0, & s_n &= \chi_{n-1}(A)\psi_n(A)r_0.\end{aligned}$$

Substituting  $z = A$  and applying each polynomial identity to  $r_0$  gives

$$\begin{aligned}t_{n+1} &= t_n - \eta_n y_n - \xi_n A t_n, \\ t_n &= \chi_n(A)\phi_n(A)r_0 - \mu_n A p_n, \\ y_n &= \chi_{n-1}(A)\phi_n(A)r_0 - t_n - \mu_n A s_n, \\ p_{n+1} &= t_{n+1} - \tau_n \eta_n s_n + \tau_n (1 + \eta_n) p_n - \tau_n \xi_n A p_n, \\ \chi_n(A)\psi_{n+1}(A)r_0 &= t_n + \tau_n p_n.\end{aligned}$$

To make the recurrence closed, denote

$$u_n = \chi_n(A)\phi_n(A)r_0, \quad v_n = \chi_{n-1}(A)\phi_n(A)r_0.$$

Then

$$t_n = u_n - \mu_n A p_n, \quad y_n = v_n - t_n - \mu_n A s_n.$$

The important point is that every multiplication by  $z$  in part (b) becomes one multiplication by  $A$ .

**(d) Updating the approximate solution.** The BCG update is

$$x_{n+1}^{BCG} = x_n^{BCG} + \mu_n \psi_n(A)r_0.$$

The smoothed residual polynomial corresponding to the sequence  $\chi_{n+1}\phi_{n+1}$  is

$$r_{n+1}^S = \chi_{n+1}(A)\phi_{n+1}(A)r_0.$$

Because  $\chi_{n+1}(0) = 1$ , there exists an approximate solution  $x_{n+1}$  such that

$$r_{n+1}^S = b - Ax_{n+1}.$$

Using the recurrence of  $\chi_{n+1}$  and the definitions of  $t_n, y_n, p_n, s_n$ , we have

$$r_{n+1}^S = t_n - \eta_n y_n - \xi_n A t_n.$$

Since  $A t_n = A(b - A\tilde{x}_n)$  corresponds to a correction by  $\xi_n t_n$ , the solution update has the form

$$x_{n+1} = x_n + \xi_n t_n + \eta_n (x_n - x_{n-1})$$

for the pure  $\chi$ -smoothing part. Incorporating the BCG correction  $\mu_n p_n$  gives the practical update

$$x_{n+1} = x_n + \mu_n p_n + \xi_n t_n + \eta_n (x_n - x_{n-1})$$

where  $t_n, p_n, y_n, s_n$  are the auxiliary vectors generated above. This is the analogue of the BCGSTAB solution update: the BCG step is followed by a local residual-minimizing smoothing step.

**(e) Formula for  $\mu_n$  and  $\tau_n$ .** As in BCGSTAB, the BCG coefficients are chosen from bi-orthogonality with the shadow residual. Let

$$\rho_n = \langle \hat{r}_0, r_n \rangle, \quad r_n = \phi_n(A)r_0, \quad p_n^{BCG} = \psi_n(A)r_0.$$

Then

$$\mu_n = \frac{\langle \hat{r}_0, r_n \rangle}{\langle \hat{r}_0, A p_n^{BCG} \rangle}$$

and

$$\tau_n = \frac{\langle \hat{r}_0, r_{n+1} \rangle}{\langle \hat{r}_0, r_n \rangle} \frac{\mu_n}{\mu_{n-1}}$$

under the common BCGSTAB notation where  $\tau_n$  is the analogue of  $\beta_n$  in the search-direction recurrence. In terms of the smoothed vectors, the same formula is evaluated by replacing  $r_n$  and  $p_n^{BCG}$  by their available polynomial representatives before multiplication by  $\chi_n(A)$ ; for example,

$$\langle \hat{r}_0, \chi_n(A)r_n \rangle = \langle \chi_n(A)^* \hat{r}_0, r_n \rangle,$$

so the same scalar can be computed using the stored shadow sequence. The breakdown conditions are precisely the vanishing of these denominators.

**Problem 4.** Compare iterative methods for non-Hermitian matrices in the three settings of the assignment.

**Problem 5.** Let  $M = LU$  be a preconditioner for  $A$ . Show that the left, right, and split preconditioned matrices have the same eigenvalues. Does this imply the corresponding iterations converge in the same or roughly the same number of steps?

**Solution.** The three common preconditioned matrices are

$$A_L = M^{-1}A, \quad A_R = AM^{-1}, \quad A_S = L^{-1}AU^{-1}.$$

First,  $M^{-1}A$  and  $AM^{-1}$  are similar whenever  $M$  is nonsingular:

$$AM^{-1} = M(M^{-1}A)M^{-1}.$$

Hence they have exactly the same eigenvalues.

For the split form, since  $M = LU$ , we have

$$M^{-1}A = U^{-1}L^{-1}A.$$

The split matrix is

$$L^{-1}AU^{-1}.$$

These two matrices are similar:

$$L^{-1}AU^{-1} = U(U^{-1}L^{-1}A)U^{-1} = U(M^{-1}A)U^{-1}.$$

Therefore

$$\sigma(M^{-1}A) = \sigma(AM^{-1}) = \sigma(L^{-1}AU^{-1}).$$

However, equal eigenvalues do not imply identical convergence behavior for nonnormal matrices. Krylov convergence depends not only on eigenvalues, but also on eigenvectors, field of values, conditioning of the eigenvector matrix, and the residual norm in which optimality is measured.

Thus:

- (a) They do not necessarily converge in exactly the same number of steps in finite precision. In exact arithmetic, unrestarted Krylov methods may terminate in the same degree if the minimal polynomial is the same, but practical convergence histories can differ.
- (b) They do not necessarily converge in roughly the same number of steps for arbitrary matrices, especially for highly nonnormal matrices.
- (c) The most accurate statement is that they may behave similarly when the transformations are well conditioned and the matrix is not strongly nonnormal. For ill-conditioned or highly nonnormal problems, convergence can be substantially different.

**Problem 6.** Let  $A$  and its preconditioner  $M$  be Hermitian positive definite. Since  $M^{-1}A$  is self-adjoint with respect to the  $A$ -inner product, write a left-preconditioned CG-like algorithm for

$$M^{-1}Ax = M^{-1}b$$

using the  $A$ -inner product. The algorithm should use only one matrix-vector product with  $A$  per CG step.

**Solution.** Define

$$\langle u, v \rangle_A = \langle Au, v \rangle.$$

For  $B = M^{-1}A$ , we verify self-adjointness in the  $A$ -inner product:

$$\langle Bu, v \rangle_A = \langle AM^{-1}Au, v \rangle = \langle u, AM^{-1}Av \rangle = \langle u, Bv \rangle_A,$$

because  $A$  and  $M$  are Hermitian positive definite. Also

$$\langle Bu, u \rangle_A = \langle AM^{-1}Au, u \rangle > 0$$

for  $u \neq 0$ . Therefore CG can be applied to  $Bx = M^{-1}b$  in the  $A$ -inner product.

A convenient implementation is the following. Let

$$r_k = b - Ax_k, \quad z_k = M^{-1}r_k.$$

The preconditioned residual for  $M^{-1}Ax = M^{-1}b$  is  $z_k$ . The algorithm is:

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**Algorithm 1**  $A$ -inner-product left-preconditioned CG

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- 1: Choose  $x_0$ ; compute  $r_0 = b - Ax_0$  and solve  $Mz_0 = r_0$ .
  - 2: Set  $p_0 = z_0$ .
  - 3: **for**  $k = 0, 1, 2, \dots$  until convergence **do**
  - 4:   Compute  $w_k = Ap_k$ .  $\triangleright$  the only multiplication by  $A$  in this step
  - 5:    $\alpha_k = \frac{\langle r_k, z_k \rangle}{\langle p_k, w_k \rangle}$ .
  - 6:    $x_{k+1} = x_k + \alpha_k p_k$ .
  - 7:    $r_{k+1} = r_k - \alpha_k w_k$ .
  - 8:   Solve  $Mz_{k+1} = r_{k+1}$ .
  - 9:    $\beta_k = \frac{\langle r_{k+1}, z_{k+1} \rangle}{\langle r_k, z_k \rangle}$ .
  - 10:    $p_{k+1} = z_{k+1} + \beta_k p_k$ .
  - 11: **end for**
- 

Although this looks like standard preconditioned CG, its derivation is from applying CG to  $M^{-1}A$  under the  $A$ -inner product. The formula uses only one  $A$ -multiplication per iteration because  $Ap_k$  is reused for both the step length and the residual update.

**Problem 7.** Describe right and split preconditioned CGNR/CGNE, and the two centered normal-equation versions.

**Solution.** Recall the two normal-equation formulations:

$$\text{CGNR: } A^*Ax = A^*b, \quad \text{CGNE: } AA^*y = b, \quad x = A^*y.$$

(a) **Four right/split preconditioned versions.**

**Right P-CGNR.** Let  $x = M^{-1}u$ . Then

$$AM^{-1}u = b.$$

Applying CGNR gives

$$(AM^{-1})^*(AM^{-1})u = (AM^{-1})^*b,$$

that is

$$M^{-*}A^*AM^{-1}u = M^{-*}A^*b, \quad x = M^{-1}u.$$

**Right P-CGNE.** Again set  $x = M^{-1}u$ . CGNE for  $AM^{-1}u = b$  is

$$(AM^{-1})(AM^{-1})^*y = b, \quad u = (AM^{-1})^*y.$$

Thus

$$AM^{-1}M^{-*}A^*y = b, \quad x = M^{-1}u = M^{-1}M^{-*}A^*y.$$

**Split P-CGNR.** Let  $M = LU$  and set  $x = U^{-1}u$ . The split-preconditioned system is

$$L^{-1}AU^{-1}u = L^{-1}b.$$

CGNR gives

$$(L^{-1}AU^{-1})^*(L^{-1}AU^{-1})u = (L^{-1}AU^{-1})^*L^{-1}b.$$

Equivalently,

$$U^{-*}A^*L^{-*}L^{-1}AU^{-1}u = U^{-*}A^*L^{-*}L^{-1}b, \quad x = U^{-1}u.$$

**Split P-CGNE.** For

$$L^{-1}AU^{-1}u = L^{-1}b,$$

CGNE gives

$$(L^{-1}AU^{-1})(L^{-1}AU^{-1})^*y = L^{-1}b, \quad u = (L^{-1}AU^{-1})^*y.$$

Thus

$$L^{-1}AU^{-1}U^{-*}A^*L^{-*}y = L^{-1}b,$$

and

$$x = U^{-1}u = U^{-1}U^{-*}A^*L^{-*}y.$$

Suitable weighted inner products can be used so that the coefficient matrices are explicitly Hermitian positive definite in the chosen inner product.

**(b) Centered versions.**

**Centered NE.** The system is

$$AM^{-1}A^*y = b, \quad x = M^{-1}A^*y.$$

Since  $M$  is Hermitian positive definite,  $AM^{-1}A^*$  is Hermitian positive semidefinite, and positive definite if  $A$  is nonsingular. CG can be applied directly:

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**Algorithm 2** Centered NE-CG

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- 1: Choose  $y_0$ ; set  $x_0 = M^{-1}A^*y_0$  and  $r_0 = b - AM^{-1}A^*y_0$ .
  - 2:  $p_0 = r_0$ .
  - 3: **for**  $k = 0, 1, 2, \dots$  **do**
  - 4:    $q_k = AM^{-1}A^*p_k$ .
  - 5:    $\alpha_k = (r_k^*r_k)/(p_k^*q_k)$ .
  - 6:    $y_{k+1} = y_k + \alpha_k p_k$ .
  - 7:    $r_{k+1} = r_k - \alpha_k q_k$ .
  - 8:    $\beta_k = (r_{k+1}^*r_{k+1})/(r_k^*r_k)$ .
  - 9:    $p_{k+1} = r_{k+1} + \beta_k p_k$ .
  - 10: **end for**
  - 11: Return  $x = M^{-1}A^*y$ .
- 

**Centered NR.** The system is

$$A^*M^{-1}Ax = A^*M^{-1}b.$$

Apply CG directly to this Hermitian positive definite system:

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**Algorithm 3** Centered NR-CG

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- 1: Choose  $x_0$ ; set  $r_0 = A^*M^{-1}(b - Ax_0)$ .
  - 2:  $p_0 = r_0$ .
  - 3: **for**  $k = 0, 1, 2, \dots$  **do**
  - 4:    $q_k = A^*M^{-1}Ap_k$ .
  - 5:    $\alpha_k = (r_k^*r_k)/(p_k^*q_k)$ .
  - 6:    $x_{k+1} = x_k + \alpha_k p_k$ .
  - 7:    $r_{k+1} = r_k - \alpha_k q_k$ .
  - 8:    $\beta_k = (r_{k+1}^*r_{k+1})/(r_k^*r_k)$ .
  - 9:    $p_{k+1} = r_{k+1} + \beta_k p_k$ .
  - 10: **end for**
- 

**Problem 8.** Assume a Hermitian positive definite matrix  $M$  is used to precondition GMRES for a non-Hermitian system. Give a formal description of left-preconditioned GMRES with the  $M$ -inner product, including modified Gram-Schmidt. What optimality property holds? What if  $A$  is also Hermitian? What is a potential advantage?

**Solution.** Consider the left-preconditioned system

$$M^{-1}Ax = M^{-1}b.$$

Define the  $M$ -inner product and norm by

$$\langle u, v \rangle_M = v^*Mu, \quad \|u\|_M = \sqrt{\langle u, u \rangle_M}.$$

Let

$$B = M^{-1}A, \quad c = M^{-1}b.$$

GMRES with the  $M$ -inner product constructs an  $M$ -orthonormal basis

$$Q_n = [q_1, \dots, q_n], \quad Q_n^*MQ_n = I.$$

Because inner products require  $Mq_i$ , we store both  $q_i$  and  $z_i = Mq_i$ .

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**Algorithm 4** Left-preconditioned GMRES with  $M$ -inner product

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- 1: Choose  $x_0$  and compute  $r_0 = c - Bx_0 = M^{-1}(b - Ax_0)$ .
- 2: Compute  $z = Mr_0 = b - Ax_0$ .
- 3:  $\beta = \|r_0\|_M = \sqrt{r_0^*Mr_0} = \sqrt{r_0^*(b - Ax_0)}$ .
- 4:  $q_1 = r_0/\beta$  and  $z_1 = Mq_1 = z/\beta$ .
- 5: **for**  $j = 1, 2, \dots, n$  **do**
- 6:   Compute  $w = Bq_j = M^{-1}Aq_j$  by first forming  $Aq_j$  and then solving with  $M$ .
- 7:   Store  $s = Mw = Aq_j$ .
- 8:   **for**  $i = 1, 2, \dots, j$  **do** ▷ modified Gram-Schmidt in  $M$ -inner product
- 9:      $h_{ij} = q_i^*Mw = q_i^*s = z_i^*w$ .
- 10:     $w = w - h_{ij}q_i$ .
- 11:     $s = s - h_{ij}z_i$ . ▷ maintain  $s = Mw$
- 12:   **end for**
- 13:    $h_{j+1,j} = \sqrt{w^*s}$ .
- 14:    $q_{j+1} = w/h_{j+1,j}$ ,    $z_{j+1} = s/h_{j+1,j}$ .
- 15: **end for**
- 16: Solve the least-squares problem

$$y_n = \arg \min_y \|\beta e_1 - \bar{H}_n y\|_2.$$

- 17: Return  $x_n = x_0 + Q_n y_n$ .
- 

The Arnoldi relation is

$$BQ_n = Q_{n+1}\bar{H}_n, \quad Q_{n+1}^*MQ_{n+1} = I.$$

Therefore

$$\begin{aligned} \|c - B(x_0 + Q_n y)\|_M &= \|r_0 - BQ_n y\|_M \\ &= \|Q_{n+1}(\beta e_1 - \bar{H}_n y)\|_M \\ &= \|\beta e_1 - \bar{H}_n y\|_2. \end{aligned}$$

Thus the computed  $x_n$  satisfies the optimality property

$$\boxed{x_n = \arg \min_{x \in x_0 + \mathcal{K}_n(M^{-1}A, r_0)} \|M^{-1}(b - Ax)\|_M}$$

or equivalently

$$\|M^{-1}(b - Ax)\|_M^2 = (b - Ax)^* M^{-1}(b - Ax).$$

If  $A$  is also Hermitian, then  $M^{-1}A$  is self-adjoint with respect to the  $M$ -inner product:

$$\langle M^{-1}Au, v \rangle_M = v^* Au = \langle u, M^{-1}Av \rangle_M.$$

Hence the Arnoldi process reduces to a three-term Lanczos process in the  $M$ -inner product. The resulting method is therefore a minimum-residual method for a self-adjoint preconditioned operator. A potential advantage is that one can exploit short recurrences, reducing storage and orthogonalization cost compared with full GMRES, while still measuring residuals in a norm naturally induced by the preconditioner.